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An Explainable Deep Learning Framework for Forest Wildfire Classification Using Custom CNNs and Transfer Learning

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***Abstract*—Forest wildfire detection has become increasingly important due to the rising frequency and severity of wildfires driven by climate change, prolonged droughts, and land-use changes, making early and accurate detection essential for reducing fire spread and mitigating losses. Deep learning has significantly advanced vision-based wildfire detection. However, existing studies often evaluate models under different datasets, preprocessing pipelines, and experimental settings, making fair comparison difficult. Moreover, explainability is rarely incorporated to systematically assess how different models make classification decisions. To address these challenges, this study presents a unified framework for binary forest wildfire image classification by evaluating a hyperparameter-optimized custom convolutional neural network (CNN), trained using both Adam and SGD optimizers, alongside three transfer learning models: MobileNetV2, ResNet50V2, and EfficientNetB3. The proposed framework employs a standardized preprocessing pipeline, progressive data augmentation, automated hyperparameter optimization using Keras Tuner, and gradient-based saliency mapping to improve model interpretability. Experimental results demonstrate that ResNet50V2 achieved the highest performance, attaining 98% accuracy and an AUC-ROC of 1.000, while the optimized custom CNN matched or outperformed MobileNetV2 and EfficientNetB3 despite being trained from scratch. These findings demonstrate the effectiveness of systematic hyperparameter optimization and explainability for developing accurate, transparent, and trustworthy vision-based wildfire detection systems.**

***Index Terms*—Wildfire detection, CNN, Binary classification, Transfer learning, Custom CNN, Data augmentation, Grad-CAM, Explainable AI.**

I. Introduction

Wildfires are among the most destructive natural hazards, causing severe environmental, economic, and societal damage. Rising temperatures, prolonged droughts, and land-use changes have increased wildfire frequency and intensity worldwide, making early and accurate detection essential for reducing fire spread and mitigating losses [3], [17]. Traditional detection methods, including lookout towers, human patrols, and satellite monitoring, suffer from limitations such as delayed response, limited temporal resolution, and sensitivity to environmental conditions [18]. Vision-based systems using surveillance cameras or UAVs provide a promising alternative. However, conventional image-processing techniques based on handcrafted features often perform poorly under varying illumination, smoke appearance, and complex backgrounds [5].

Recent advances in deep learning have significantly improved image classification and object detection across many application domains. Convolutional neural networks (CNNs) have demonstrated strong performance for wildfire detection by automatically extracting discriminative visual features from fire and smoke images [2], [3], [9], [10], [13], [14]. Furthermore, transfer learning approaches based on pretrained architectures such as MobileNetV2, ResNet50V2, and EfficientNet have achieved high classification accuracy while reducing training requirements [3]–[7], [11].

Despite these advances, direct comparison among existing wildfire detection models remains challenging because studies often employ different datasets, image preprocessing strategies, experimental settings, and evaluation metrics. Consequently, it is difficult to determine which architectures provide the best balance between accuracy, computational efficiency, and generalization capability. Moreover, although explainable artificial intelligence (XAI) methods, such as gradient-based saliency visualization, have been introduced to interpret deep learning predictions, their application for systematically comparing the learned visual representations and decision-making behavior of different wildfire detection models remains limited [8].

Recent work spans several distinct problems that are often grouped together. One line forecasts fire danger rather than classifying imagery: Prapas *et al.* [19] predict daily wildfire

danger from environmental variables, Zhang *et al.* [20] model global susceptibility, and Maggioros and Tsalkitzis [21] apply transfer learning to danger prediction. A second line addresses image-based detection with pretrained backbones: Khan *et al.* [23] introduce the DeepFire dataset together with a transfer-learning benchmark, Park *et al.* [22] treat the task as multilabel classification for response decision support, and Yandouzi *et al.* [24] and Mahmoud *et al.* [26] report accuracies above 95% while optimising for inference cost. A third line extends beyond CNNs, with Ghali *et al.* [7] evaluating transformer backbones for UAV imagery and Sathishkumar *et al.* [25] addressing catastrophic forgetting when a detector is updated with new fire data. Each establishes a result within its own experimental setting, but none holds dataset, preprocessing, training schedule and evaluation protocol fixed while varying only the architecture, which is what a like-for-like comparison requires.

To address the identified research gap, this study develops and evaluates a wildfire image classification framework by comparing a custom CNN with three widely adopted transfer learning architectures: MobileNetV2, ResNet50V2, and EfficientNetB3. All models are evaluated using the same dataset, standardized preprocessing and data augmentation procedures, and identical evaluation metrics to ensure a fair and reliable comparison. Furthermore, gradient-based saliency mapping is incorporated to visualize model predictions and analyze whether the networks focus on meaningful wildfire-related regions, thereby improving model transparency and trustworthiness.

II. Research Methodology

This section describes the proposed methodology, as illustrated in Fig. 1 including dataset preparation, preprocessing, augmentation, model development, hyperparameter optimization, training, and evaluation procedures.

A. Dataset Description

The dataset used in this study was the *Forest Fire Images* dataset obtained from Kaggle [1]. It consists of labeled images belonging to two classes: *Fire* and *Non-Fire*, making it suitable for binary image classification. The images were collected from diverse real-world environments, including dense forests, open vegetation areas, and woodland regions near urban settings. Consequently, the dataset exhibits substantial variation in fire intensity, smoke patterns, illumination conditions, weather effects, and background complexity.

TABLE I: Dataset composition before and after cleaning.

Class	Original	Removed	Cleaned
Fire	2500	388	2112
Non-Fire	2500	1	2499
Total	**5000**	**389**	**4611**

B. Class Distribution and Data Quality

The original training set contained 5,000 images, equally divided between the *Fire* and *Non-Fire* classes. During data cleaning, a PIL-based verification process identified 389 corrupted images that could not be successfully opened or converted to RGB format. No corrupted images were found in the test set. Image corruption was almost entirely concentrated in the *Fire* class, resulting in a cleaned dataset containing 2,112 *Fire* images and 2,499 *Non-Fire* images. This introduced a moderate class imbalance, as summarized in Table I.

C. Preprocessing Pipeline

Following data cleaning, all images were resized to 128×128 pixels and pixel values were normalized to the range $[0, 1]$. A supplementary quality assessment was applied to flag degraded frames. Images whose mean grey-level fell below 10 were marked near-blank, and images whose Laplacian variance fell below 50 were marked excessively blurred. The scan flagged 679 of the 5,000 original training images (13.6%). These images were deliberately retained. Low-contrast and motion-blurred frames are characteristic of the conditions a deployed wildfire detector operates under—smoke-obscured scenes, dusk and night-time imagery, and hand-held or UAV-mounted cameras—so excluding them would make the evaluation easier than the target application and bias the reported performance upward. The two criteria are also coarse heuristics rather than defect detectors: a dense smoke plume legitimately produces a low Laplacian variance without being a corrupted image. Only the 389 files that failed PIL decoding entirely (Table I) were removed.

To mitigate the class imbalance introduced during cleaning, inverse-frequency class weights were computed and applied during training. The four remaining models receive these normalised inputs directly. EfficientNetB3 is a special case. The Keras implementation places `Rescaling`$(1/255)$ and `Normalization` as the first internal layers of the backbone, and its `preprocess_input` function is a documented no-op; the model therefore expects unscaled $[0, 255]$ inputs. Supplying inputs already normalised to $[0, 1]$ applies the division by 255 a second time, so the backbone receives values in approximately $[0, 0.0039]$ before ImageNet mean–variance normalisation. The initial experiments in this study supplied $[0, 1]$ inputs to all five models for pipeline consistency, which for EfficientNetB3 was a preprocessing fault rather than a fairness safeguard. EfficientNetB3's results should therefore be read as a lower bound on its achievable performance under this pipeline, and its position in the ranking is not evidence about the architecture itself.

D. Data Augmentation: Three Versions

To evaluate the impact of augmentation, three augmentation regimes were compared. Version A served as a baseline without augmentation. Version B applied geometric transformations, including horizontal flipping, rotation up to $\pm 20°$, zoom up to 15%, and width/height shifts up to 10%. Version C extended these transformations with brightness adjustments between 70–130% and channel shifts of up to 20 intensity levels to simulate realistic variations in outdoor wildfire imagery.

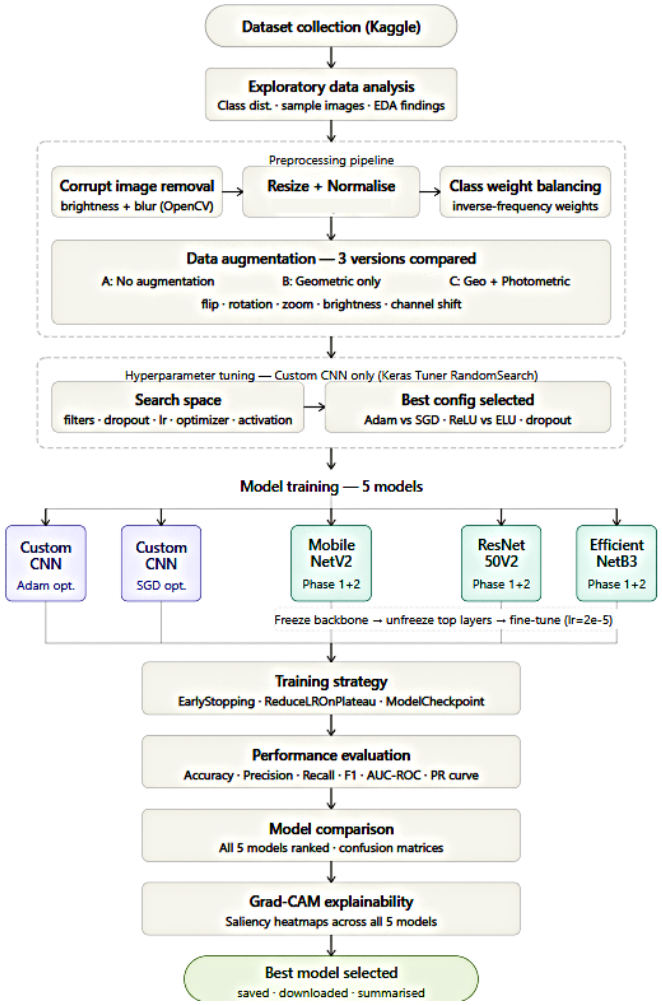


Fig. 1: End-to-End Preprocessing, Training, and Evaluation Pipeline.

Fig. 2 presents representative examples of the augmented images. The best-performing configuration (Version C) was subsequently used for all final model training.

E. Data splitting

The cleaned dataset was divided using an 80/20 stratified train-validation split, resulting in 3,689 training images and 922 validation images while preserving class proportions. The independent test set contained 50 images, evenly distributed between the two classes (25 *Fire* and 25 *Non-Fire*).

The independent test set of 50 images is small, and its implications for the reported metrics are discussed in Section IV.

F. Hyperparameter Tuning

To optimize the Custom CNN architecture, automated hyperparameter tuning was performed using Keras Tuner's `RandomSearch` algorithm. Validation accuracy was used as the optimization objective over 10 search trials. The search space included the number of filters in the first convolutional layer (32 or 64), dropout rate (0.2, 0.4, or 0.5), learning rate



Fig. 2: Representative training samples before (left) and after (right) augmentation. The panels show different source images.

(1×10^{-2}, 1×10^{-3}, or 1×10^{-4}), optimizer (Adam or SGD), and activation function (ReLU or ELU).

The best-performing configuration employed ReLU activation, a dropout rate of 0.2, a learning rate of 1×10^{-4}, the Adam optimizer, 32 convolutional filters, and an L2 regularization coefficient of 0.001, achieving a validation accuracy of 94.8%. Although effective, the limited number of trials means that the search should be viewed as exploratory rather than exhaustive. Future work could investigate larger search spaces and Bayesian optimization techniques.

G. Model Architectures

Five deep learning models were developed and evaluated. The first two models were custom convolutional neural networks trained from scratch using the architecture obtained from hyperparameter tuning. The Custom CNN (Adam) was trained using the Adam optimizer, while the Custom CNN (SGD) used SGD with Nesterov momentum at a learning rate of 1×10^{-3}. Since both models share an identical architecture, their comparison isolates the effect of optimizer choice. The remaining models employed transfer learning using ImageNet-pretrained backbones. A two-stage training strategy was adopted in which the backbone was initially frozen while the classification head was trained, followed by selective fine-tuning of deeper layers using a reduced learning rate. MobileNetV2 was fine-tuned from layer 100 and represents a lightweight architecture suitable for resource-constrained environments. ResNet50V2 was fine-tuned from layer 150 and leverages residual connections to facilitate deeper feature learning. EfficientNetB3, a compound-scaled architecture, was fine-tuned from layer 200. Table II summarizes the key architectural and training configurations of all models.

TABLE II: Architecture and training configuration summary.

Model	Total Params	FT Layer	Opt.	LR
CNN (Adam)	489,153	Scratch	Adam	1e-4
CNN (SGD)	489,153	Scratch	SGD	1e-3
MobileNetV2	2,618,945	100	Adam	2e-5
ResNet50V2	24,123,393	150	Adam	2e-5
EfficientNetB3	11,211,056	200	Adam	2e-5

H. Training Strategy

To ensure a fair comparison, all models were trained using a consistent optimization framework. Early stopping was employed with a patience of five epochs, restoring the model weights corresponding to the best validation performance. This reduced overfitting and prevented unnecessary training once convergence had been reached. Model optimization was further supported by the `ReduceLROnPlateau` scheduler, which reduced the learning rate by a factor of 0.5 after three epochs without validation improvement, down to a minimum value of 1×10^{-7}. In addition, model checkpointing was used to save the weights associated with the lowest validation loss achieved during training. Consequently, all reported results correspond to the best-performing version of each model.

I. Model Performance Metrics

As suggested by author [16], in this study too, models' performance was assessed on the held-out test set using Accuracy, Precision, Recall, F1-Score and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which is reported as the threshold-free measure of class separability. These metrics provide complementary perspectives on classification performance and class discrimination capability. To improve model interpretability, gradient-based saliency heatmaps [27] were generated for all five models. These visualizations highlight image regions that most strongly influenced model predictions and provide qualitative insight into the learned decision-making process. The approach follows the broader class-activation and gradient-based explainability framework introduced by Selvaraju et al. [12].

III. Results and Discussion

A. Performance Outcomes of Model Evaluation

Table III summarize the performance of all five evaluated models on the test set. Fig. 4a shows the evaluation metrics, while Fig. 4b presents the ROC and Precision–Recall curves. From these, it can be observed that among the architectures assessed, ResNet50V2 achieved the best overall performance, with an accuracy of 98%, perfect precision (1.000), an F1-score of 0.980, and perfect AUC-ROC and PR-AUC values of 1.000. These results demonstrate an excellent ability to distinguish between the *Fire* and *Non-Fire* classes under the conditions represented in the test dataset. ResNet50V2's pre-activation residual architecture and two-stage fine-tuning schedule are the most plausible contributors to this result, although the test set is too small to attribute the margin to any single design choice. Its AUC-ROC of 1.000 indicates near-perfect separability of the two classes across decision thresholds on this test set. Furthermore, two custom CNN models demonstrated strong performance despite being trained entirely from scratch. Both the Adam- and SGD-optimized variants achieved an accuracy of 96% and an AUC-ROC of 0.989. Notably, the Adam-based model achieved perfect recall (1.000), indicating that no fire images were missed, while the SGD-based model provided a more balanced trade-off between precision and recall. These findings suggest that a carefully designed custom architecture can perform competitively with transfer-learning-based approaches. For binary wildfire image classification on a moderately sized and well-curated dataset, pretrained feature extractors may not be strictly necessary to achieve high predictive performance.

TABLE III: Test set performance of all five models ($n = 50$).

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	PR-AUC
Custom CNN (Adam)	0.96	0.926	1.000	0.962	0.989	0.987
Custom CNN (SGD)	0.96	0.960	0.960	0.960	0.989	0.990
MobileNetV2 TL	0.86	0.950	0.760	0.844	0.990	0.990
ResNet50V2 TL	**0.98**	**1.000**	0.960	**0.980**	**1.000**	**1.000**
EfficientNetB3 TL	0.76	0.810	0.680	0.739	0.851	0.861

Treating each classification as an independent Bernoulli trial, 95% Wilson score intervals were computed for each model's test-set accuracy [15] (Table IV).

TABLE IV: 95% Wilson score confidence intervals for test-set accuracy ($n = 50$).

Model	Accuracy	95% CI
Custom CNN (Adam)	0.96	[0.865, 0.989]
Custom CNN (SGD)	0.96	[0.865, 0.989]
MobileNetV2 TL	0.86	[0.738, 0.930]
ResNet50V2 TL	0.98	[0.895, 0.996]
EfficientNetB3 TL	0.76	[0.626, 0.857]

The intervals are relatively wide because of the limited test set size and show substantial overlap among the top-performing models. In particular, ResNet50V2's interval ([0.895, 0.996]) overlaps with the intervals of both custom CNN variants ([0.865, 0.989]), indicating that the observed 2-percentage-point accuracy difference is not statistically distinguishable at this sample size. In contrast, the confidence intervals for MobileNetV2 and EfficientNetB3 lie noticeably lower, providing stronger evidence that these models underperformed relative to the leading architectures. Overall, the results suggest that ResNet50V2 was the strongest model in this study, although a larger and more diverse test set would be required to confirm the ranking with greater statistical confidence.

B. Results of Optimizer Comparison: Adam vs. SGD

The two custom CNN variants achieved identical overall accuracy (96%) and AUC-ROC (0.989), indicating that optimizer choice had limited impact on overall classification performance for this architecture. However, differences emerged in the precision–recall trade-off. The Adam-based model achieved perfect recall (1.000), successfully identifying all fire images, whereas the SGD variant achieved balanced precision and recall values of 0.960. From an operational perspective, the Adam model may be preferable for wildfire detection, where missed fire events generally carry greater

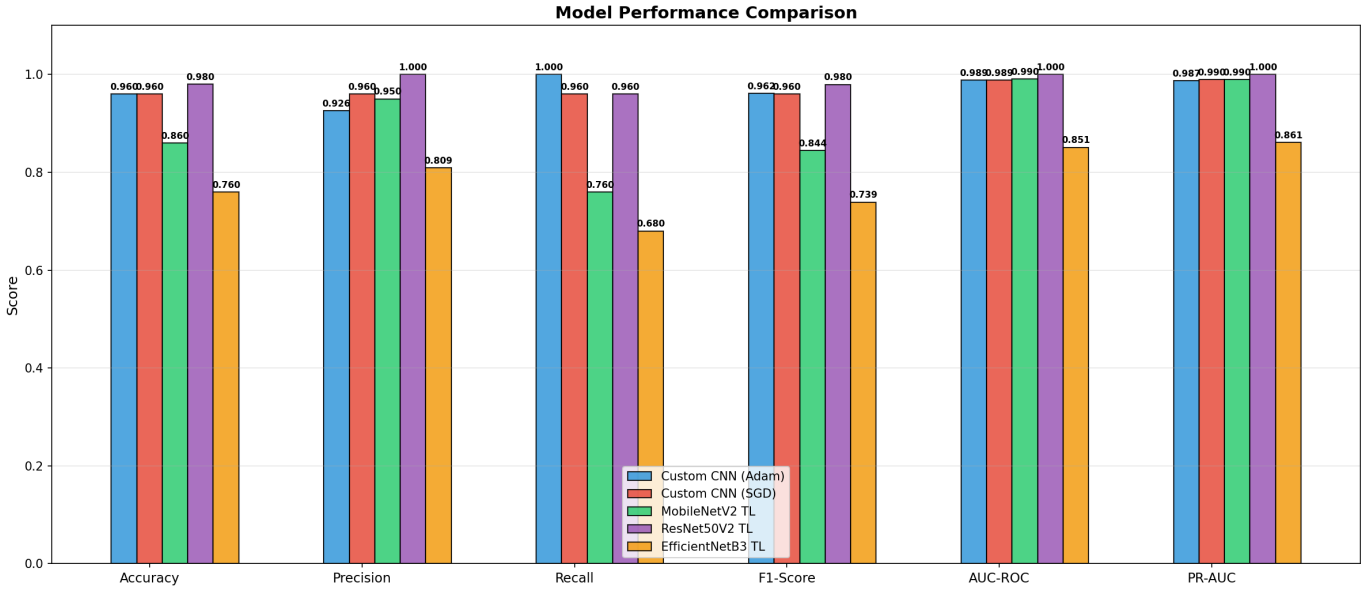


Fig. 3: Performance Comparison Across All Five Models.

consequences than false alarms. Nevertheless, since only a single architecture was evaluated with both optimizers, these findings should be interpreted as architecture-specific rather than broadly generalizable.

C. Ablation Study Results for Data Augmentation

To assess the contribution of data augmentation, three augmentation strategies were evaluated using the Custom CNN (Adam). The results are presented in Table V.

Training without augmentation (Version A) achieved an accuracy of 88% and an F1-score of 0.885. Introducing geometric transformations (Version B) increased accuracy to 94% and improved the F1-score to 0.939, demonstrating the benefit of exposing the model to variations in image orientation, scale, and positioning. Extending the pipeline with photometric transformations (Version C) further increased accuracy to 96% and achieved perfect recall (1.000). The progressive improvement across versions demonstrates the importance of training data diversity. Geometric augmentation improved robustness to changes in viewpoint and framing, while photometric augmentation enhanced generalization to varying illumination and environmental conditions.The increase from 88% to 96% accuracy identifies data augmentation as the most influential single component of the training pipeline examined here, although the comparison between individual augmentation versions rests on one run each.

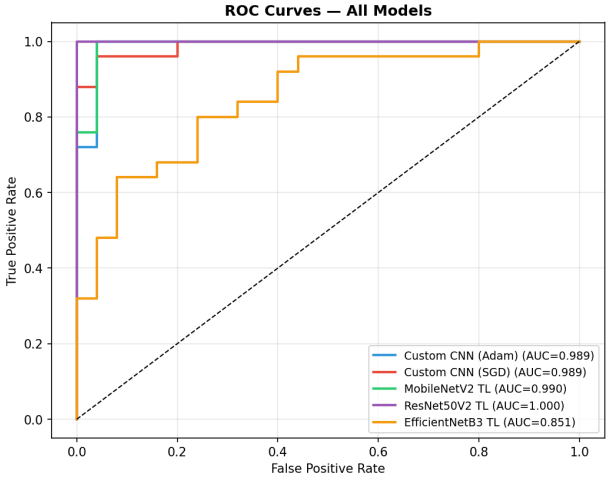
TABLE V: Augmentation ablation study results on the Custom CNN models with Adam optimization.

Augmentation Version	Accuracy	Precision	Recall	F1-Score
Version A — None	0.88	0.852	0.920	0.885
Version B — Geometric Only	0.94	0.958	0.920	0.939
Version C — Geometric + Photometric	0.96	0.926	1.000	0.962

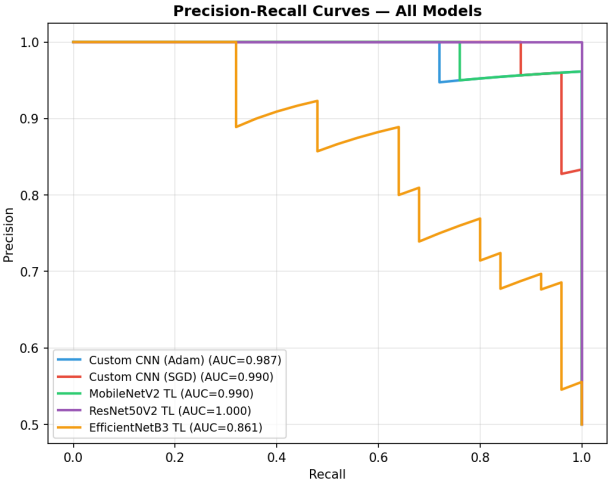
Each augmentation regime was evaluated using a single training run, so variability arising from random initialisation was not quantified. This constrains which differences in Table V can be interpreted. The eight-point accuracy gain from Version A to Version C is large relative to the two-point resolution of a 50-image test set and is reported here as a trend. The smaller step between adjacent versions corresponds to one or two test images and cannot be separated from run-to-run variation without repeated seeds. Repeating each configuration across multiple random initialisations would be required to establish the ordering of B and C with confidence.

D. Comparative Evaluation of Transfer Learning Models

While ResNet50V2 benefited substantially from transfer learning, the remaining pretrained models exhibited contrasting behavior. *MobileNetV2* achieved a high AUC-ROC (0.990) despite its lower recall (0.760), indicating that it learned a strong ranking function but produced an overly conservative classification boundary at the selected threshold. Several missed fire detections occurred in images where flames occupied only a small region of the frame or were partially obscured by smoke. In practical deployments, the classification threshold could be adjusted to prioritize recall and reduce the likelihood of missing genuine fire events. Given its lightweight architecture and strong discriminative ability, MobileNetV2 remains a promising candidate for edge-based wildfire monitoring systems. *EfficientNetB3*, by contrast, substantially underperformed relative to the other models. Of the two explanations offered in the submitted analysis, one can now be stated concretely. As described in Section II-C, the Keras EfficientNetB3 implementation rescales inputs internally and expects unscaled $[0, 255]$ values, so supplying inputs already normalised to $[0, 1]$ applied the division by 255 twice and left the backbone operating on values of approximately



(a) ROC curves for all five models.



(b) Precision–Recall curves for all five models.

Fig. 4: Performance comparison of all evaluated models on the test set.



Fig. 5: The Obtained Prediction Samples Using Custom CNN (Adam).

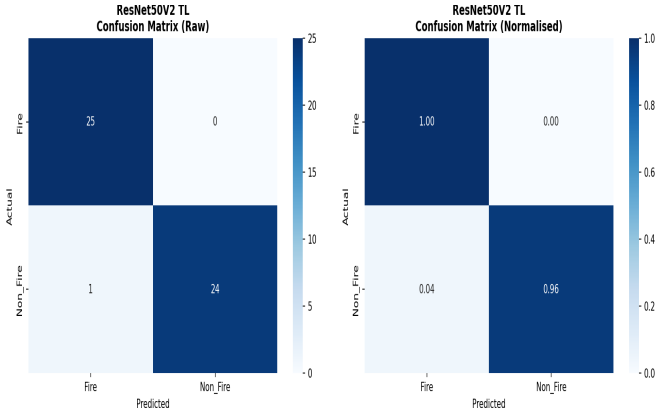


Fig. 6: The Obtained Confusion Matrix using ResNet50V2 Model.

$[0, 0.004]$. This is a defect in the input pipeline rather than a property of the architecture, and it is the more likely of the two explanations. The second factor—that compound-scaled architectures require more training data than the approximately 3,689 images available here—remains plausible but untested. Isolating the two effects would require retraining EfficientNetB3 under both input ranges and across a range of training-set sizes, which was outside the scope of this study. EfficientNetB3's results should therefore be read as a lower bound on its achievable performance under this pipeline, and its position in the ranking is not evidence about the architecture itself.

E. Saliency-Based Explainability

To better understand model decision-making, gradient-based saliency maps were generated for all five architectures (Fig. 9).

Across the best-performing models, including ResNet50V2 and both custom CNN variants, activations consistently focused on flame regions, smoke plumes, and characteristic orange-red color gradients. This suggests that the models learned semantically meaningful wildfire features rather than relying primarily on background information. ResNet50V2 produced the most spatially localized activations, with attention concentrated directly on fire and smoke regions. The custom CNN models displayed slightly broader activation patterns but remained strongly centered on fire-related visual cues. In contrast, EfficientNetB3 frequently generated diffuse activations extending into background areas, consistent with its weaker classification performance.

Analysis of misclassified images provided additional insight. The Custom CNN (Adam) false-positive case showed strong activation over warm-colored vegetation rather than



Fig. 7: The Obtained Predictions using ResNet50V2 Model.



Fig. 8: The Obtained Prediction Samples using EfficientNetB3.

flame regions, indicating that the model occasionally over-relied on color-based cues. MobileNetV2's false negatives generally occurred when fire occupied only a small portion of the image or was partially obscured by smoke, producing weak and scattered activations that reflected uncertainty rather than confident misclassification. EfficientNetB3 frequently failed to localize fire regions even in visually obvious wildfire scenes, suggesting that its errors were driven by poor feature localization rather than ambiguity in the images themselves.

IV. Conclusions and Future Work

This paper presented a methodology for automated forest wildfire image classification using five deep learning models, including a custom convolutional neural networks trained from scratch and transfer-learning approaches based on MobileNetV2, ResNet50V2, and EfficientNetB3. All models were evaluated under a unified experimental framework using a publicly available Kaggle dataset. Experimental results demonstrated that ResNet50V2 achieved the best overall performance, obtaining 98% accuracy, an F1-score of 0.980, and an AUC-ROC of 1.000 on the test set. The hyperparameter-optimized custom CNN also achieved competitive results, reaching 96% accuracy without relying on pretrained weights, highlighting its potential for domain-specific wildfire detection and resource-constrained applications. The augmentation ablation showed that combining geometric and photometric transformations improved test-set accuracy by eight percentage points over no augmentation, and saliency analysis indicated that the higher-performing models responded to fire-related image regions rather than incidental background features.

These findings hold under the conditions of this study: a single publicly available dataset, a 50-image independent test set, and one training run per configuration. At this test-set size a single misclassified image shifts accuracy by two percentage points, and the 95% confidence intervals for ResNet50V2 and the two custom CNN variants overlap, so the ranking among the leading models is a point-estimate trend rather than a statistically confirmed ordering. The results are therefore best read as evidence that a hyperparameter-optimised custom CNN is competitive with pretrained backbones on a curated wildfire dataset of this scale, rather than as a general recommendation of one architecture over another. Confirming the ordering, and establishing generalisation to operational imagery, would require a substantially larger and more diverse independent test set drawn from more than one source. Future work will evaluate the framework on larger multi-source wildfire datasets, extend it to video-based detection, optimise models for edge deployment, and incorporate uncertainty estimation to complement point-estimate metrics.

Acknowledgement

During this project, we used ChatGPT (OpenAI) solely as a language assistance tool to improve grammar, clarity, and sentence structure. All technical content, analyses, and conclusions are our own work. All AI-generated suggestions were carefully reviewed and only incorporated where appropriate, without changing the technical meaning of the report.

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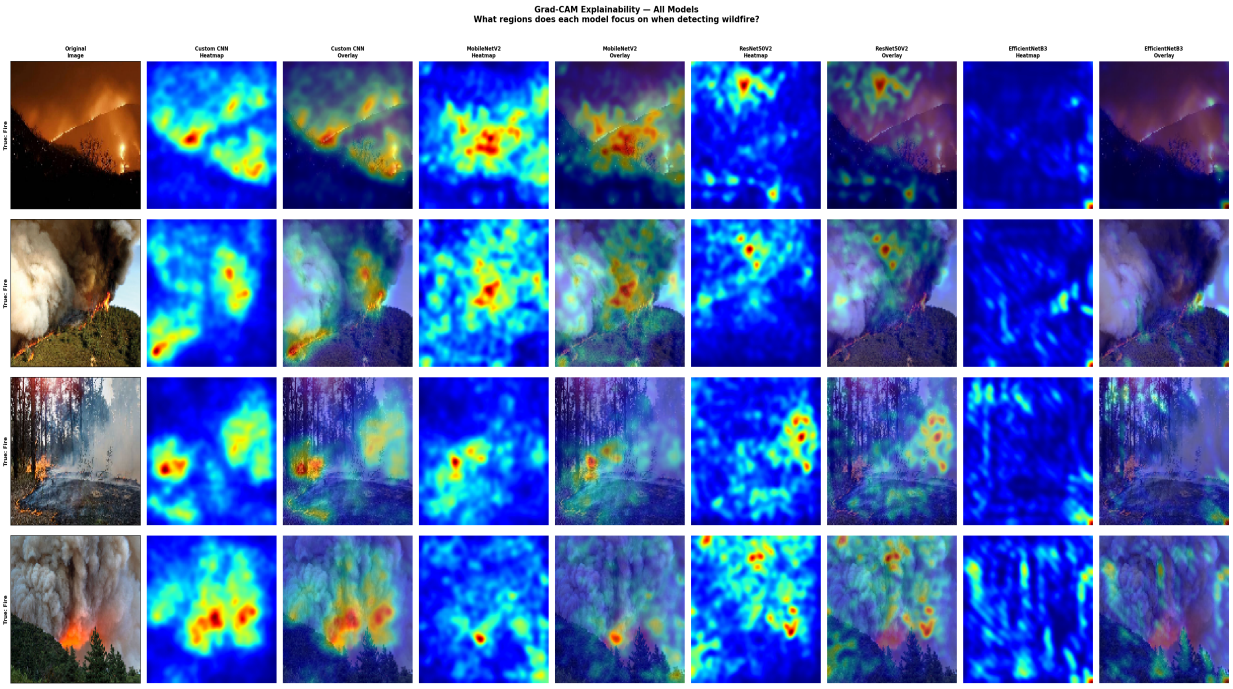


Fig. 9: The Obtained Gradient-Based Saliency Maps Across All Five Models.

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